

class notes
10/25

Topological Sort: Decrementing Functions and Loop Invariants

CSCI 432

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Name:

Who did you work with today?

Topological Sort: The Algorithm

Topological sort takes a poset $(P, \leq)$ and returns a total order of all objects that is *compatible* with the partial order of the poset. By compatible, we mean that for all $a \leq b \in P$, we have a appears before b in the total order. For example, consider P to be a set of points in $\mathbb{R}^2$ and let the partial order be the product order; that is, $(a_x, a_y) \leq (b_x, b_y)$ if and only if $a_x \leq b_x$ and $a_y \leq b_y$. Given a set:

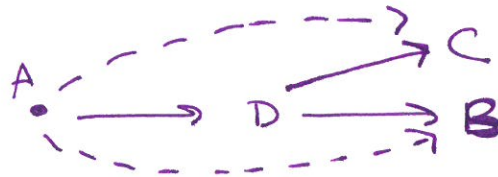
$$P = \{(0, 0), (\pi, \pi), (2, 4), (1, 2)\}$$

we can return the following total order:

$$[(0, 0), (1, 2), (2, 4), (\pi, \pi)].$$

Note that this is not the only compatible total order. What other total order could have been returned?

Answer



Hasse Diagram

The following algorithm takes as input a poset, represented as a directed acyclic graph, and returns a compatible total order. If the poset orders $a \leq b$, then the DAG has an edge from vertex a to vertex b . For this algorithm, vertices have at least one attribute: outgoing, which stores a list of vertices that define the outgoing edges.



Definitions:

- DAG = directed acyclic graph
- indegree = # edges going into a vertex
- outdegree = # edge going out of a vertex

Algorithm 1 Topological Sort

Input: $G = (V, E)$, a DAG

Output: L , a list of vertices in G in a total order compatible with the partial order of the DAG.

```
1: for  $v \in V$  do
2:   Add an attribute ' $v.count$ ' and initialize it to the indegree of  $v$ .
3: end for
4:  $Q \leftarrow$  the set of vertices with ' $count=0$ '.
5:  $L \leftarrow$  array of length  $|V|$ 
6:  $i \leftarrow 1$ 
7: while  $Q$  is not empty do
8:    $v \leftarrow Q.POP$ 
9:    $L[i] \leftarrow v$ 
10:   $i++$ 
11:  for  $u \in v.outgoing$  do
12:     $u.count--$ 
13:    if  $u.count = 0$  then
14:      Add  $u$  to  $Q$ 
15:    end if
16:  end for
17: end while
18: return  $L$ 
```

In Line 4, why is Q not empty? ie - why does there exist a vertex with indegree = 0?

[Proof by contradiction]

Explain how topological sort is helpful for dynamic programming.

Answer

Walk through a small example.

Answer

Topological Sort: The Decrementing Function

What is the decrementing function for the for loop? Line 11. v is fixed.

Answer

Let $i_j = \#$ of times we've executed the for loop for this particular v .

$$D_1: S \rightarrow \mathbb{N}$$

$$s \mapsto |V| - i_j$$

$$\left. \vphantom{\begin{matrix} D_1 \\ D_2 \end{matrix}} \right\}$$

$$D_2: S \rightarrow \mathbb{N}$$

$$state \mapsto \sum_{u \in V} u.\text{count}$$

What is the decrementing function for the while loop?

Answer

$$D: S \rightarrow \mathbb{N}$$

$$state \mapsto |V| - i$$

Topological Sort: The Loop Invariant

Here, we investigate partial correctness of the while loop. For this loop, what are the following statements (start with your best guess, then come back and revise if needed):

The loop guard G :

The post-condition Q :

The pre-condition P :

The loop invariant $L = L_i$:

Can you use this loop invariant to prove partial correctness (Remember, there are three parts to partial correctness: Initialization, Maintenance, and End).

Answer